Biagio SCARCIELLO

Embedded Software & Systems Engineer | Railway Safety-Critical Systems

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Senior Embedded Software & Systems Engineer with 8+ years of experience delivering safety-critical railway systems across high-speed, metro, and regional platforms. Expert in Embedded C/C++, real-time system design, and onboard communication protocols (TRDP, IPTCom, Ethernet). Experienced across the full engineering lifecycle: system architecture definition, integration, verification & validation, and on-site commissioning. Complemented by hands-on full-stack and mobile development — Python/FastAPI backends, React and Next.js front-ends, and Flutter applications on PostgreSQL and Docker — delivered as production systems for real clients.

Technical Skills

Embedded & Real-Time	**Embedded C/C++**, real-time systems design, memory management & performance optimization, embedded Linux, debugging & troubleshooting in field environments
Communication	**TRDP** (Train Real-time Data Protocol), **IPTCom**, Ethernet-based onboard networks, MQTT, TCP/IP, multi-vehicle communication logic, gateway design
System Engineering	System architecture definition, **IV&V**, commissioning support, requirements analysis & traceability (**DOORS**), issue tracking (Jira), safety-critical railway domain
Programming Languages	**C** (Embedded), **C++**, **Python**, JavaScript/TypeScript, Dart, MATLAB/Simulink
Web & Full-Stack	**FastAPI**, Flask, **React**, Next.js, SQLAlchemy, **PostgreSQL**, REST API design, JWT & RBAC, **Docker**
Mobile	**Flutter** (Android tablet applications), Dart, device provisioning & offline-capable clients
Development Tools	CMake, **Git**, SVN, GitHub, Wireshark, CI/CD fundamentals

Work Experience

Present – 2024
Software Lead — Diagnostic Systems, Hitachi Rail STS S.p.A., Naples, Italy
Diagnostic systems for metro, tram, and high-speed platforms

- **Software lead** for the diagnostic systems of **Rome Metro (lines A/B)** and **Florence Tram**, owning the software from development through integration and on-site commissioning
- **Software development support** on the **Milan Metro (lines 1, 2, 3)** diagnostic system, contributing to development, integration, and issue resolution
- Led a focused stabilization phase on the Rome Metro project ahead of official customer testing, **resolving 230 open issues within one month** and securing the validation milestones on schedule
- Led the full software lifecycle of onboard diagnostic systems, developing the HMI and optimizing the embedded backend framework in Embedded C
- Integrated the HMI and diagnostic logic for the **HABD** (Hot Axle Box Detector) and **DNRA** (Locked Axle Detection) safety systems on the **ETR1000 Frecciarossa**, covering sensor data handling, alarm management, and driver notification

Technologies: Embedded C, embedded Linux, Ethernet onboard networks, Jira

2024 – 2022
System Engineer — Event Recorder & Safety-Critical Subsystems, Hitachi Rail STS S.p.A., Naples, Italy
System ownership of onboard safety-critical products

- **System owner** of the **Event Recorder** product line: defined the complete system architecture and requirements, driving the solution from specification through integration, validation, and customer acceptance for the metro networks of **Milan, Rome, and Washington (WMATA)**
- System Engineer referent for Bogie Control System communication (**TCMS** integration) on the **ETR1000 Frecciarossa** high-speed platform, using message data protocol to define the communication architecture and coordinating cross-functional validation across software, hardware, and RAMS teams
- Technical referent for onboard safety-critical subsystems — Event Recorder, Dead Man, and Zero Speed detection — covering requirements analysis, traceability (DOORS), interface definition, and verification strategy
- Coordinated integration, troubleshooting, and commissioning of diagnostic subsystems in operational environments

Technologies: TRDP, IPTCom, TCMS, DOORS, system architecture, IV&V

2022 – Apr. 2018
Embedded Software Developer — Regional & High-Speed Platforms, Hitachi Rail STS S.p.A., Naples, Italy
Onboard diagnostic software development

- Developed and commissioned the diagnostic systems for the regional fleets **Trenitalia Rock**, **Trenitalia Blues**, and **Ferrovie Nord Milano**, and contributed to the high-speed platforms (*ETR1000 Frecciarossa, ETR700*)
- Developed and extended Condition-Based Maintenance (**CBM**) software in Embedded C, and designed diagnostic algorithms in C and MATLAB/Simulink
- Enhanced and refactored the embedded diagnostic software framework, improving maintainability and reuse across train platforms
- Implemented multi-vehicle composition management logic (backend and frontend) and designed diagnostic HMI components in HTML/JavaScript
- Improved the reliability and stability of distributed onboard communication networks, managing issue tracking and defect resolution through Jira

Technologies: Embedded C, MATLAB/Simulink, HTML/JavaScript, Jira

Freelance Projects

NAPP

Multi-tenant HACCP compliance platform — food safety management

napphaccp.com

Monorepo serving three actors: business owners, on-site operators, and preconfigured Android devices. Operators log temperatures, production, sanitation, and non-conformities from tablets; owners configure clients, equipment, and documents from a web panel.

- Designed the multi-tenant data model with per-client module enablement and device-level authentication (`device_id` + secret)
- Built the FastAPI backend as single source of truth, the React owner panel, the Flutter operator app, and the Next.js public landing
- Integrated **Stripe** for subscription billing and prepaid plans, handling checkout sessions and webhook-driven payment reconciliation

Technologies: FastAPI, Python 3.12, PostgreSQL, Alembic, MinIO, Stripe, React 19, TypeScript, Vite, Flutter, Next.js, Docker

CRM Evolution Time

Multi-workspace inventory platform — luxury watch trading

Inventory and sales platform for a team with distinct specializations, integrated with an external marketplace.

- Implemented role-based access control across four roles (Owner, Admin, Warehouse, Accountant) with JWT refresh-token rotation
- Built the OKTOPIA marketplace integration, Excel bulk import, and legacy data migration

Technologies: FastAPI, SQLAlchemy 2 (async), PostgreSQL, Next.js 15, TypeScript, React Query, Resend, Docker

Program Elite

Booking website & billing CRM — fitness coaching business

Two connected systems: a public Wix site handling bookings and user registrations, and a dedicated CRM that consolidates the resulting revenue into financial reporting.

- Built and maintained the **Wix** site managing bookings and user sign-ups
- Developed a dedicated CRM integrating the **Stripe** API to reconcile payments and produce consolidated financial reporting
- Automated invoice generation through **Billin**, with PDF rendering, country-specific tax handling, and delivery to clients by email via **Resend**

Technologies: Wix, FastAPI, Stripe API, Billin API, Resend, Jinja2, Tailwind CSS, PostgreSQL

Dolphin House

Booking and content website — hospitality

live demo

Full-stack website with content management, media handling, and transactional email.

- Built on Next.js with server-side rendering, Drizzle ORM over PostgreSQL, and JWT sessions
- S3-backed media pipeline with presigned uploads, and animated front-end (GSAP, Lenis)

Technologies: Next.js, React, TypeScript, Drizzle ORM, PostgreSQL, AWS S3, Resend, Tailwind CSS, Docker

Websites

Websites, e-commerce & brand design — selected client work

Design and development of responsive sites, online stores, and brand assets for local businesses and professionals, from concept to deployment.

- **The Chik** — **Shopify** e-commerce store: theme customization, product catalogue setup, and checkout configuration *(in progress)*
- **JTF Gaskets** — brand identity and product catalogue designed in **Adobe InDesign**, alongside the corporate site (jtf-gaskets.com)
- **BP Nutrition** — nutritionist practice site with service pages and contact funnel (bpnutrition.it)
- **Clinica Sabe** — clinic site with service pages and patient contact flow (sabesnc.com)
- **DV Barberia** — single-page barbershop landing site

Technologies: HTML5, CSS3, JavaScript, PHP, Shopify, Adobe InDesign, responsive design, SEO fundamentals

Education

M.Sc. **Master of Science in Automation Engineering** — University of Naples Federico II, *110/110 cum laude*

B.Sc. **Bachelor of Science in Automation Engineering** — University of Naples Federico II, *107/110*

Languages

Italian (native)

English (B2)

Interests

- CrossFit & swing dancing
- Technology & graphic design
- Travel

Publications

2019 **Model-based vehicular prognostics framework using Big Data architecture**

Computers in Industry, Elsevier

A. Petrillo, A. Picariello, S. Santini, **B. Scarciello**, G. Sperlì

- Proposed **ATOMIC** (Automotive predicTOr Maintenance In Cloud), a cloud-based Big Data infrastructure for Integrated Vehicle Health Management (IVHM) combining nonlinear discrete-time models of nominal and faulty component behaviour to estimate End-Of-Life (EOL) and Remaining Useful Life (RUL) indicators on-line
- Validated the prognostic method on the Delphi DFG1596 fuel pump and benchmarked the architecture's deployment configurations in terms of scalability and cost

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